Bacterial-nanostructure interactions

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Biophysics

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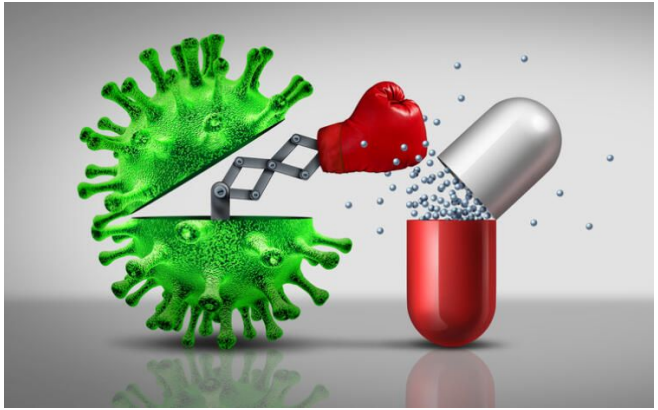
Overview

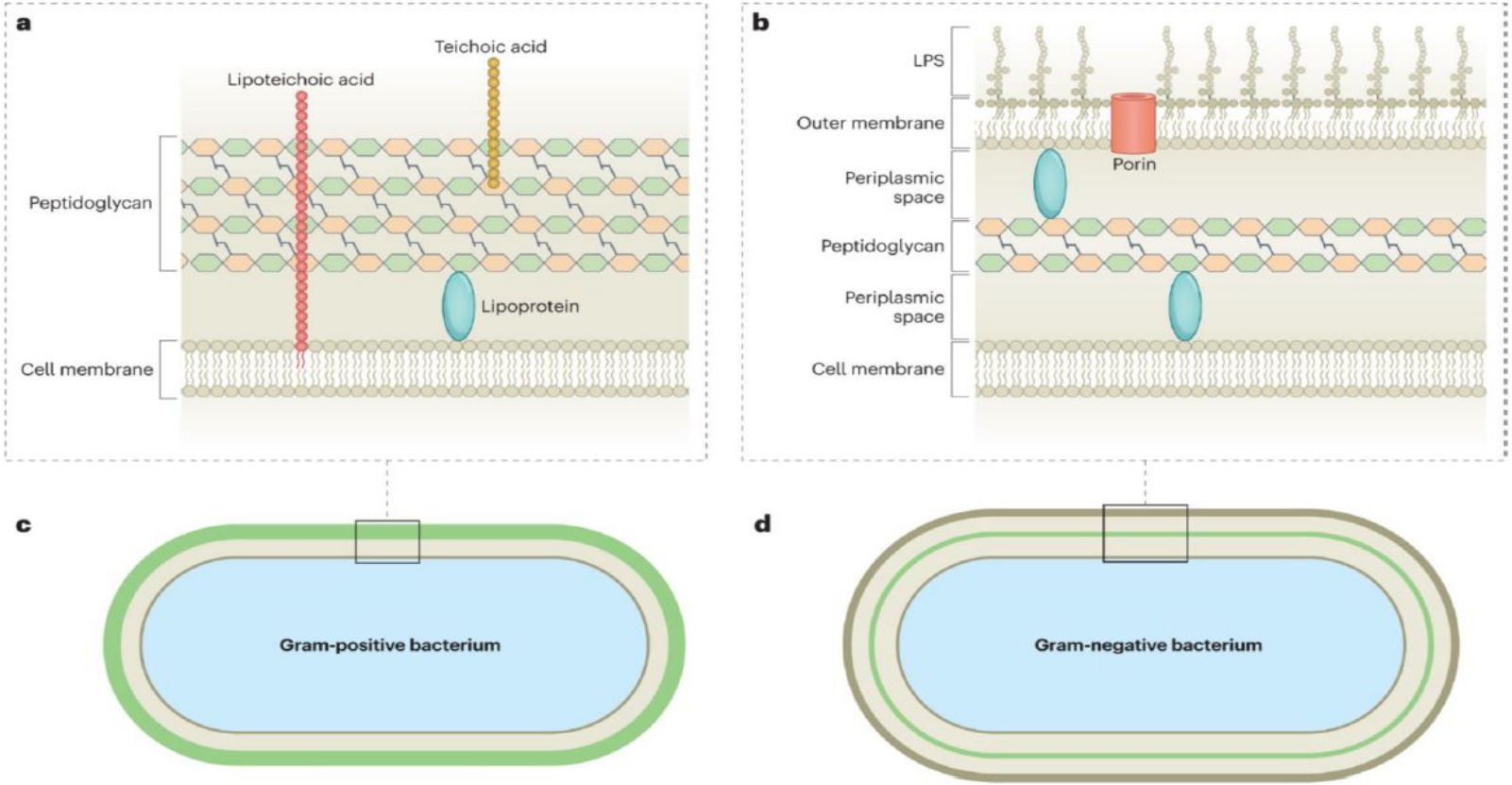
1. **Introduction**
2. **Biological framework:** The bacterium.
3. **Physical framework:** Adhesion and deformation, measurement.
4. **Biophysical modelling:** A case review.
5. **Challenges:** Knowledge gap.
6. **Outlook:** The future.
7. **Conclusion**

1. Introduction

- **Motivation**: *Antibiotic resistance is a rising global issue.*
- **Solution**: *Use surface topography to induce bacterial cell death.*
- **Starting point:** *Ivanova et al. (2012)*
- **Our focus:** *Explore state-of-the-art physical models regarding cell rupture and identify key challenges and limitations.*

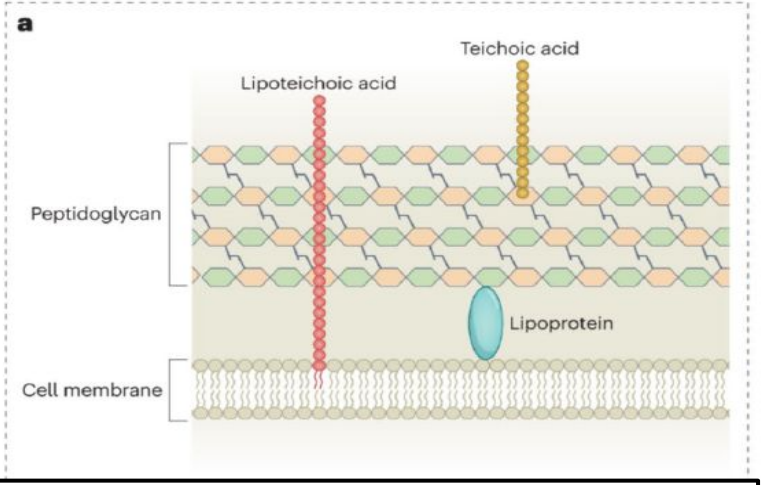


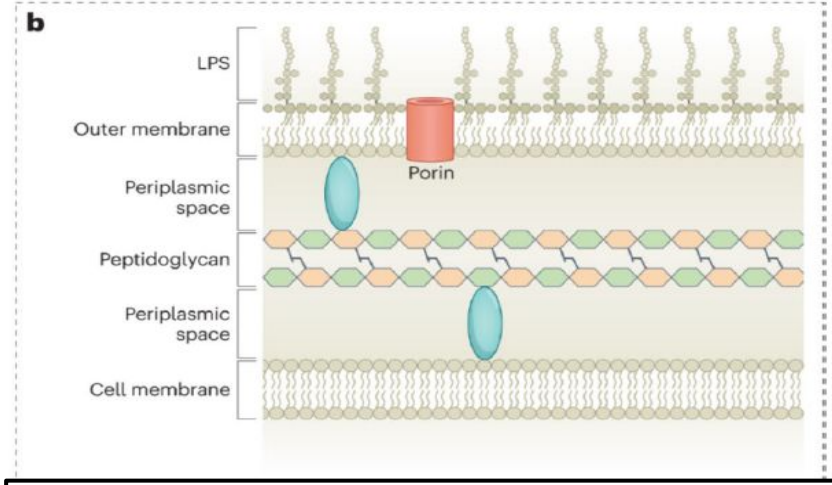
2. Biological framework: Bacterium



Wijethunge D, Mathew A, Yarlagadda PKDV. (2025).. Comprehensive review of bacterial death mechanism on nanopillared nanostructured surfaces.

2. Biological framework: Bacterium





GPB

- membrane - thick wall
- pass through porous wall (peptidoglycan)

GNB

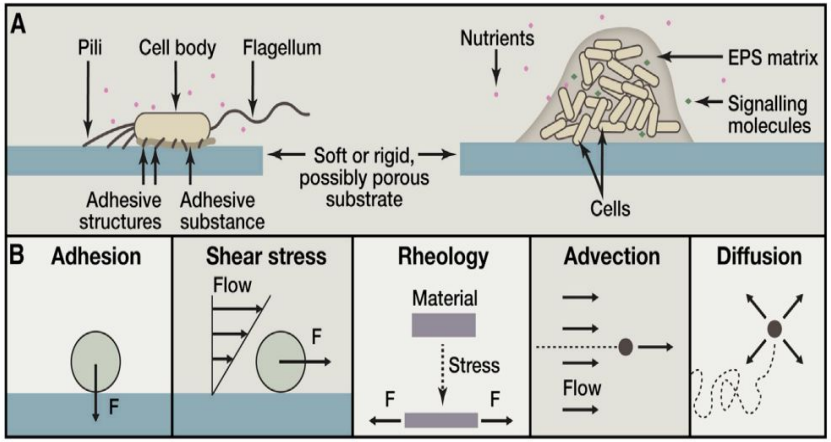
- membrane - thin wall - outer membrane
- outer membrane - chemical shield, mutations

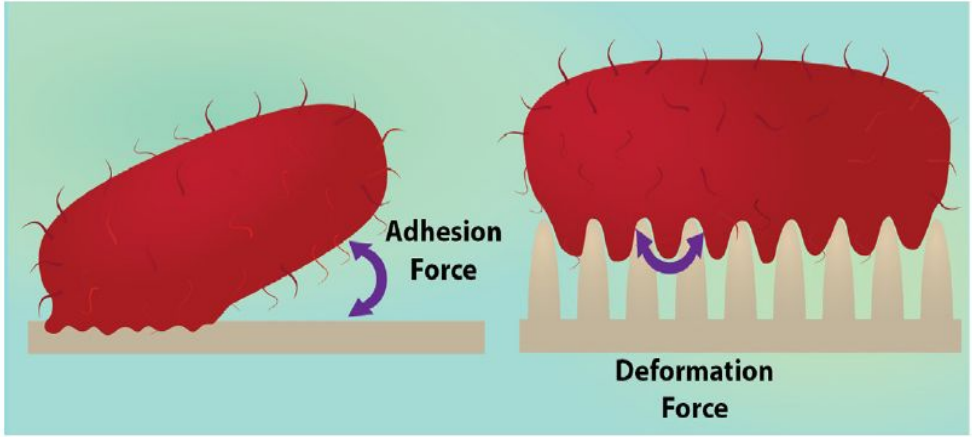
faces.

3. Physical framework: Surface interactions

adhesion = surface attachment force

deformation = applied force whose effects depend on the material's rigidity/elasticity





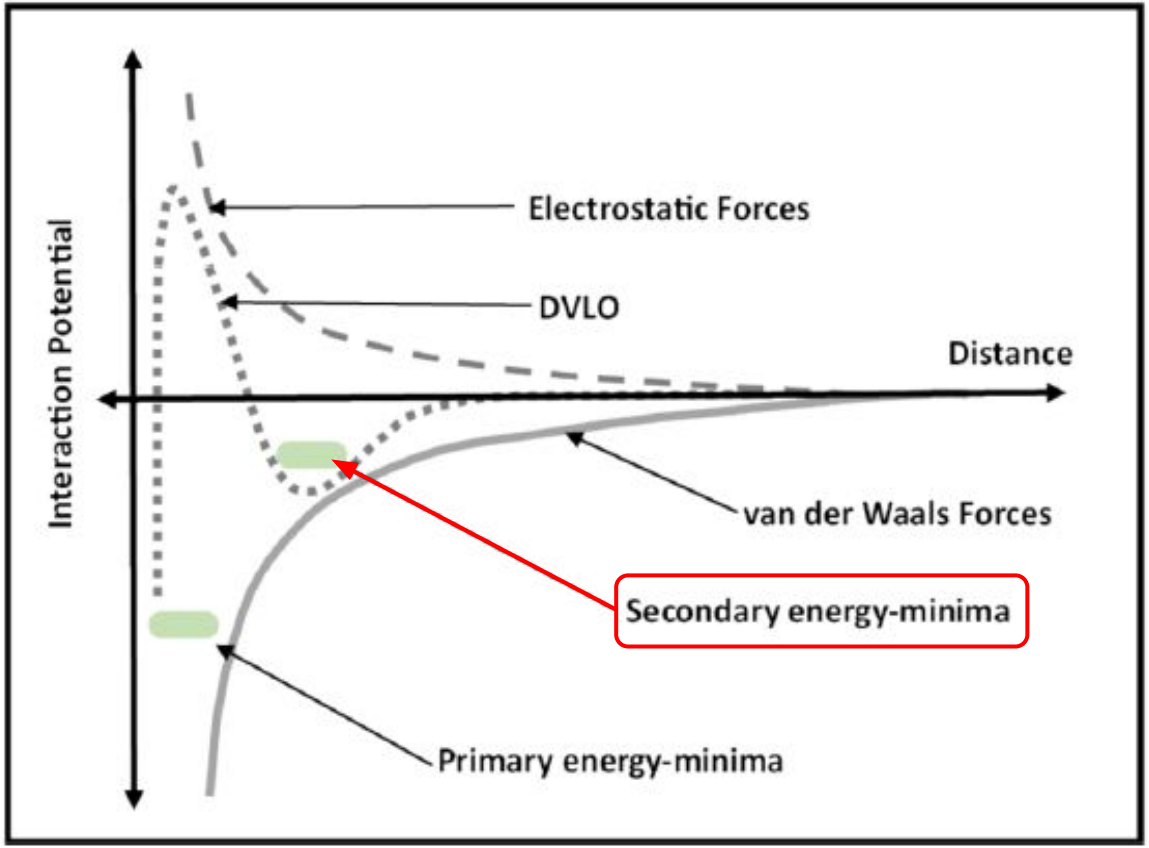
Persat A, Nadell CD, Kim MK, Ingremeau F, Siryaporn A, Drescher K, Wingreen NS, Bassler BL, Gitai Z, Stone HA. (2015). The mechanical world of bacteria.

A. Elbourne, J. Chapman, A. Gelmi, D. Cozzolino, R. J. Crawford, V. K. Truong. (2019). Bacterial-nanostructure interactions: The role of cell elasticity and adhesion forces.

3. Physical framework: Adhesion

Step 1: reversible action

- **secondary energy-minima**
- willingness to approach



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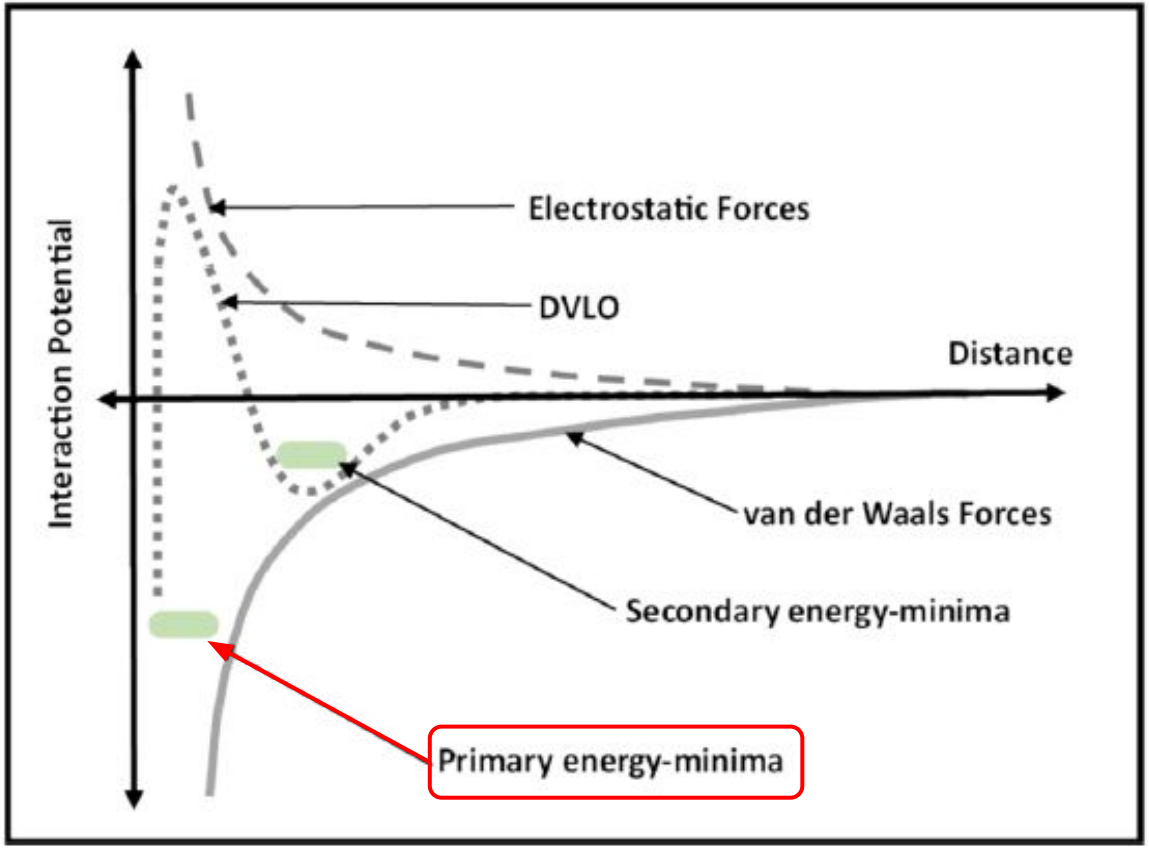
3. Physical framework: Adhesion

Step 1: reversible action

- **secondary energy-minima**
- willingness to approach

Step 2: irreversible action

- **primary energy-minima**
- contact mechanics
- *Step 3*: EPS secretion → biofilm



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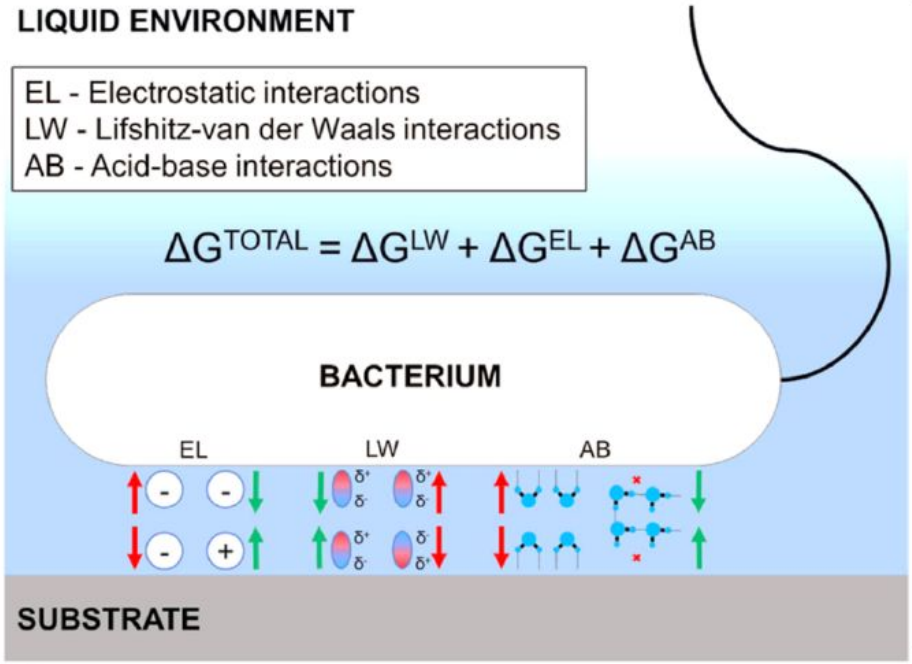
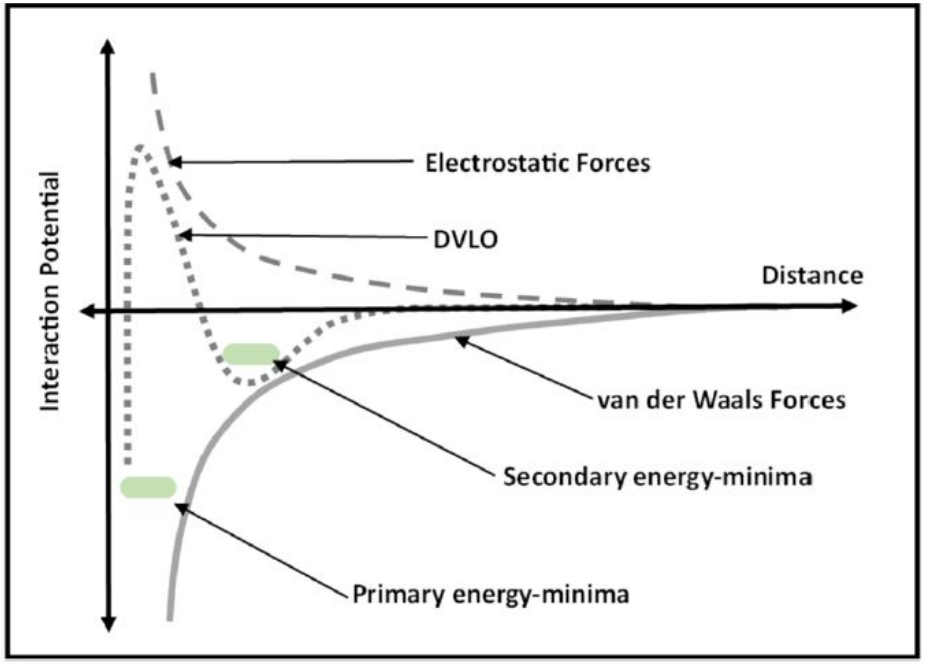
3. Physical framework: Adhesion

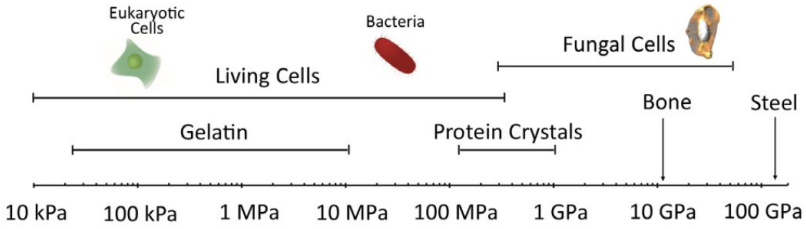




Wijethunge D, Mathew A, Yarlagadda PKDV. (2025).. Comprehensive review of bacterial death mechanism on nanopillared nanostructured surfaces.

3. Physical framework: Elasticity

Goal: Measure the elasticity of biological materials



Why? Crucial mechanical property for cell rupture

➔ without sample damage
➔ performed in-situ

HOW?

Image credit: Elbourne A et al. (2019). Bacterial-nanostructure interactions: The role of cell elasticity and adhesion forces

3. Physical framework: Elasticity

HOW?

1. Research and test available technology to perform measurements
2. Explore physical models that best fit data
3. Verify validity and improve
    a. back to steps 1 and 2

3. Physical framework: Elasticity

HOW?

1. **Research and test available technology to perform measurements**
2. Explore physical models that best fit data
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3. Physical framework: Elasticity

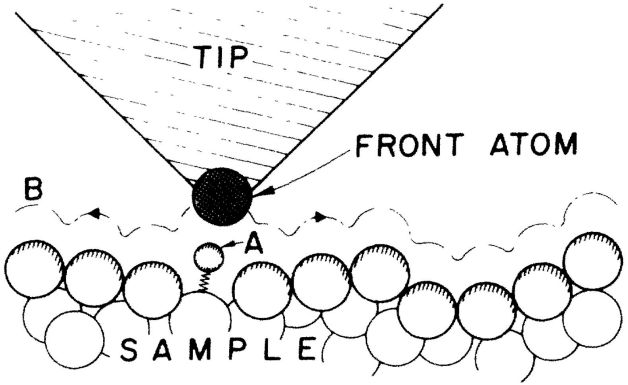
1st Step: check available technology

Atomic Force Microscopy (AFM)

- existing technology that emerged in 1986
- evolved from scanning tunneling microscope (STM)
- first purpose: material topology

➔ Advantages

- probe does not damage surface
- measures forces as small as 10^{-18} N



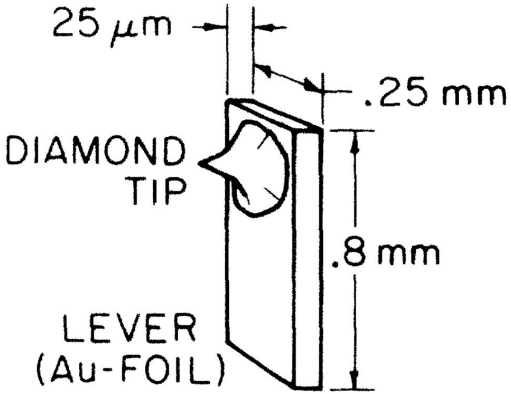
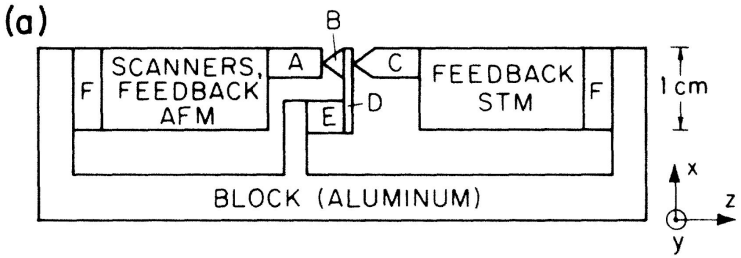


Image credit: G. Binnig and C. F. Quate (1986) Atomic Force Microscope

3. Physical framework: Elasticity

1st Step: check available technology

Atomic Force Microscopy (AFM)



➔ How it works

1. Cantilever (D) + diamond tip (B) scan the surface of the sample (A)
2. Interatomic forces between sample and tip will move cantilever in z direction
3. This will produce deformations in springs in the STM
4. Measure forces by F = -kx
5. Output of force vs indentation

Image credit: G. Binnig and C. F. Quate (1986) Atomic Force Microscope

3. Physical framework: Elasticity

HOW?

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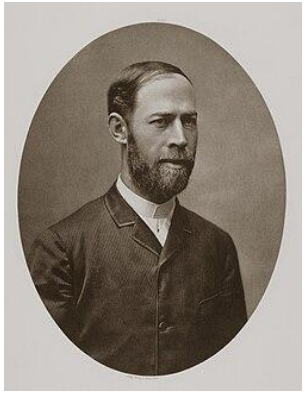
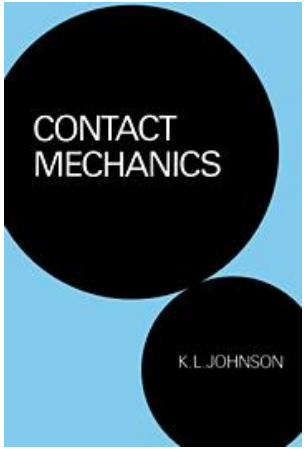
3. Physical framework: Elasticity

2nd Step: physical theory to fit data

Hertz Model

- Heinrich Hertz (1882) “On the contact of elastic solids”
  - Elliptical contact area
  - Continuous and non-conforming bodies
  - Frictionless surfaces
  - Small strains
- K. L. Johnson (1985) *Contact Mechanics*
  - removal of restrictions = progress in the field since then
  - Non-Hertzian models





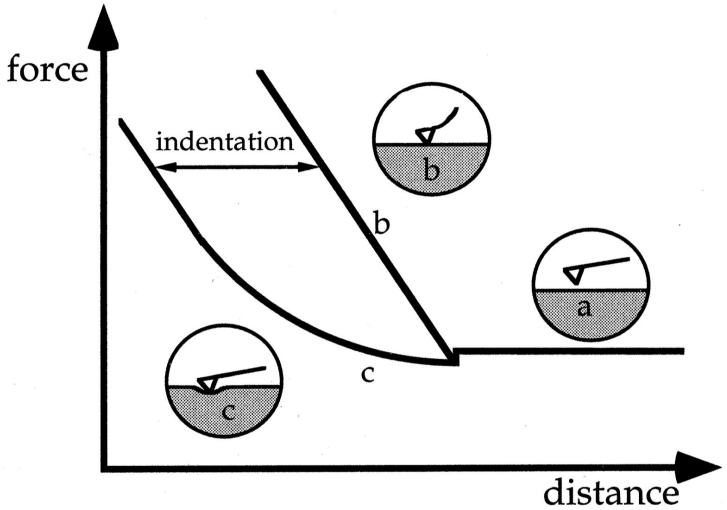
3. Physical framework: Elasticity

2nd Step: physical theory to fit data

Hertz Model: current application

Steps for probing elasticity with AFM:

1. Force curve on a hard surface as a reference
   a. ex: glass
2. Force curve on sample of interest
3. Difference of deflection = deformation of soft sample
4. Force vs indentation curve
5. Fit to $F = a(\Delta)^b$ to determine Young's modulus



$$F_{\text{cone}}(\Delta z) = \frac{2E^*}{\pi(tg(\alpha))}\Delta z^2 \qquad F_{\text{parabolic}}(\Delta z) = \frac{4\sqrt{R}}{3}E^*\Delta z^{1.5}.$$

A. Vinckier and G Semenza (1998) Measuring elasticity of biological materials by atomic force microscopy

3. Physical framework: Elasticity

2nd Step: physical theory to fit data

Johnson-Kendall-Roberts (JKR) Model

- Extension of Hertz model by Johnson (1971) - accounting for:
  - Attraction forces (van der Waals) → adhesion
  - Viscous damping
  - Spherical contact
- Models soft materials

$$F_n^{JKR} = \underbrace{\frac{4E^* a^3}{3\bar{R}}}_{\text{Hertz elastic force}} - \underbrace{\sqrt{16\pi\gamma E^* a^3}}_{\text{Adhesion term}}$$

3. Physical framework: Elasticity

2nd Step: physical theory to fit data

Derjaguin-Muller-Toporov (DMT) Model

- Similarly to JKR, also extension of Hertz model accounting for adhesion
- Models stiff materials
  - usually not the focus when dealing with cells

3. Physical framework: Elasticity

HOW?

1. Research and test available technology to perform measurements
2. Explore physical models that best fit data
3. **Verify validity and improve**
    a. back to steps 1 and 2

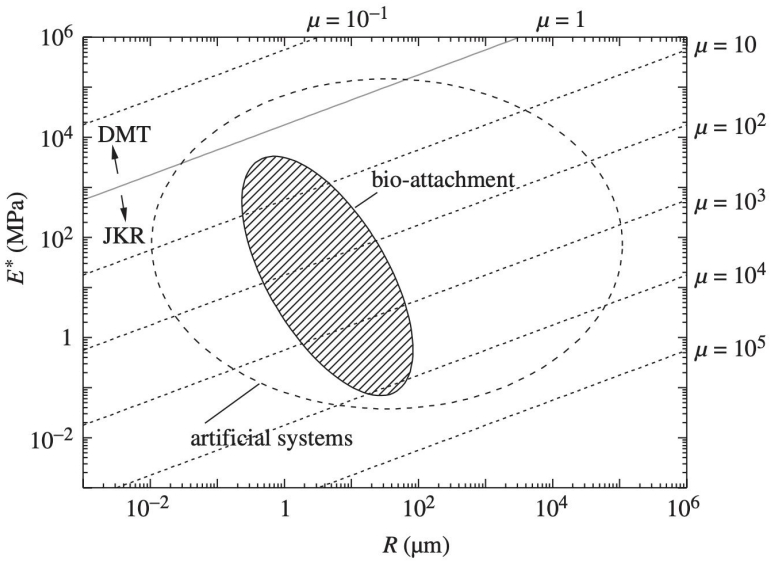
3. Physical framework: Elasticity

3rd Step: verify validity

Challenges and limitations

- Hertz theory is oversimplified
  - absence of attractive forces makes a big impact on final result
- JKR and DMT fill that gap but are still shape-restricted and limited by Tarbor's parameter

$$\mu = \frac{R^{1/3}\gamma^{2/3}}{(E^*)^{2/3}\varepsilon},$$



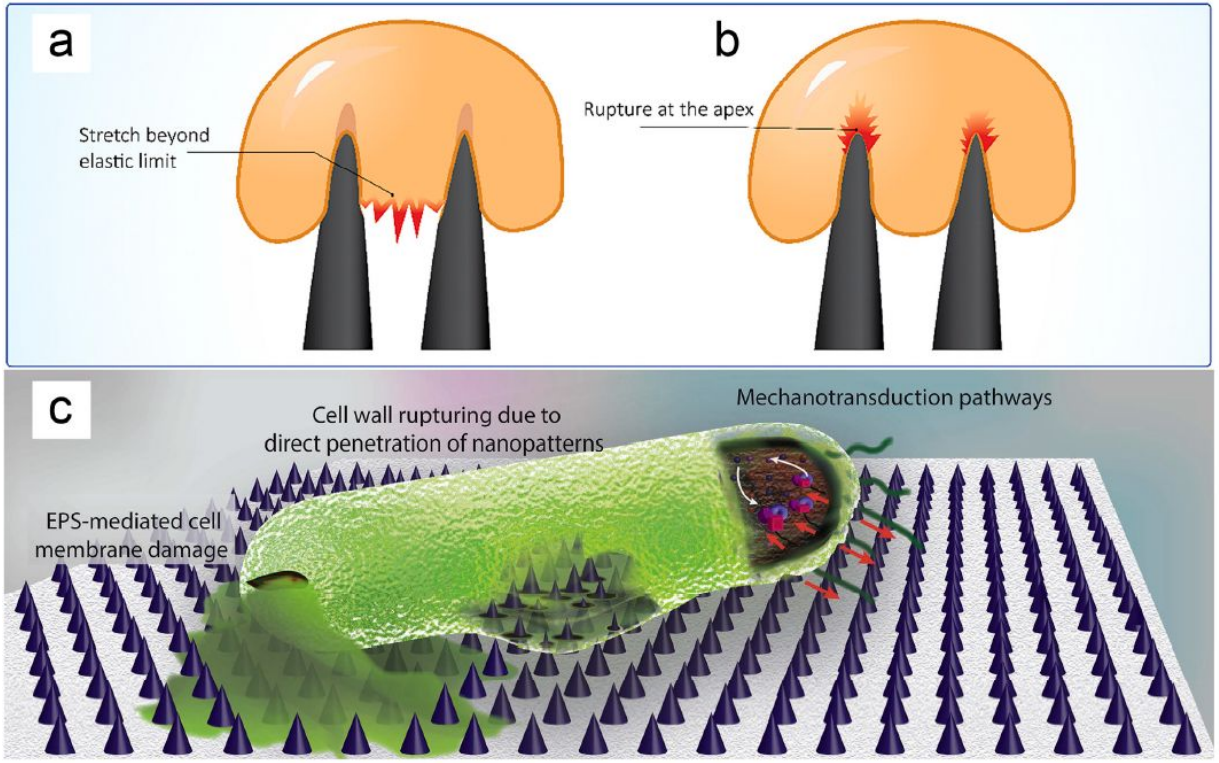
R: radius of contact sphere
γ: work of adhesion
E*: reduced Young's modulus
ε: interatomic distance at closest contact point

R. Spolenak et al. (2005) Effects of contact shape on the scaling of biological attachments

4. Biophysical modelling:

- Pogodin (2013): stretching beyond elastic limit

- **does not correlate with experimental data**



Hayles, A.; Hasan, J.; Bright, R.; Palms, D.; Brown, T.; Barker, D.; Vasilev, K. (2021). Hydrothermally etched titanium: A review on a promising mechano-bactericidal surface for implant applications.

4. Biophysical modelling: Velic et al. 2021

- a robust computational approach

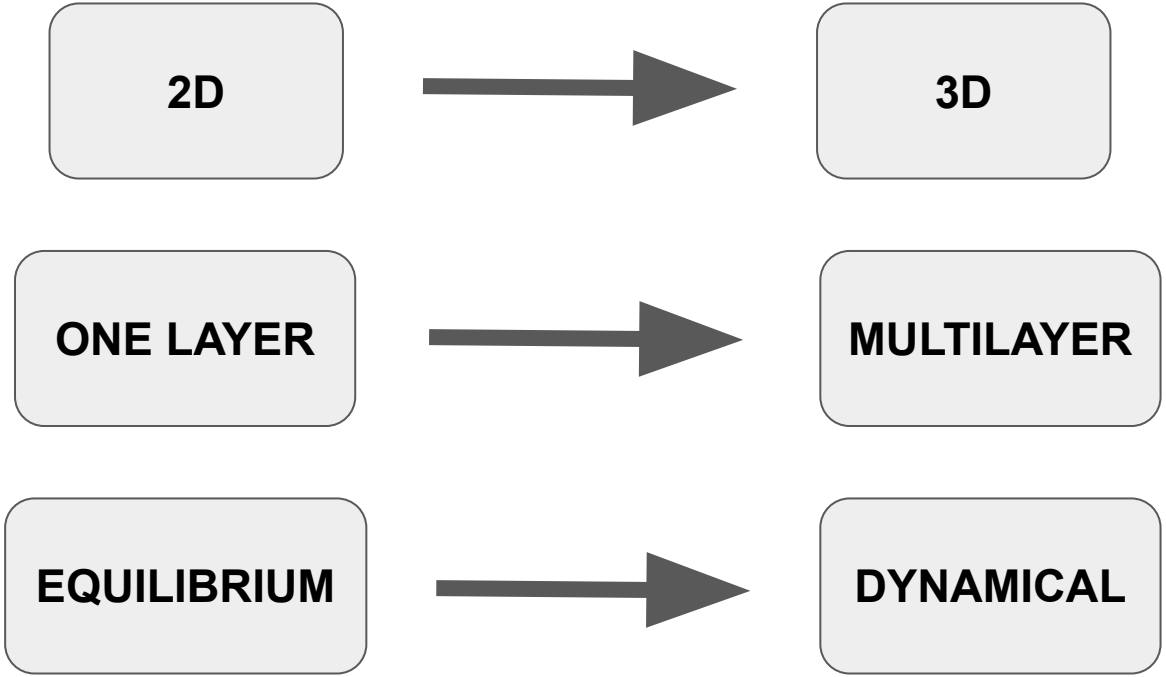
2D

ONE LAYER

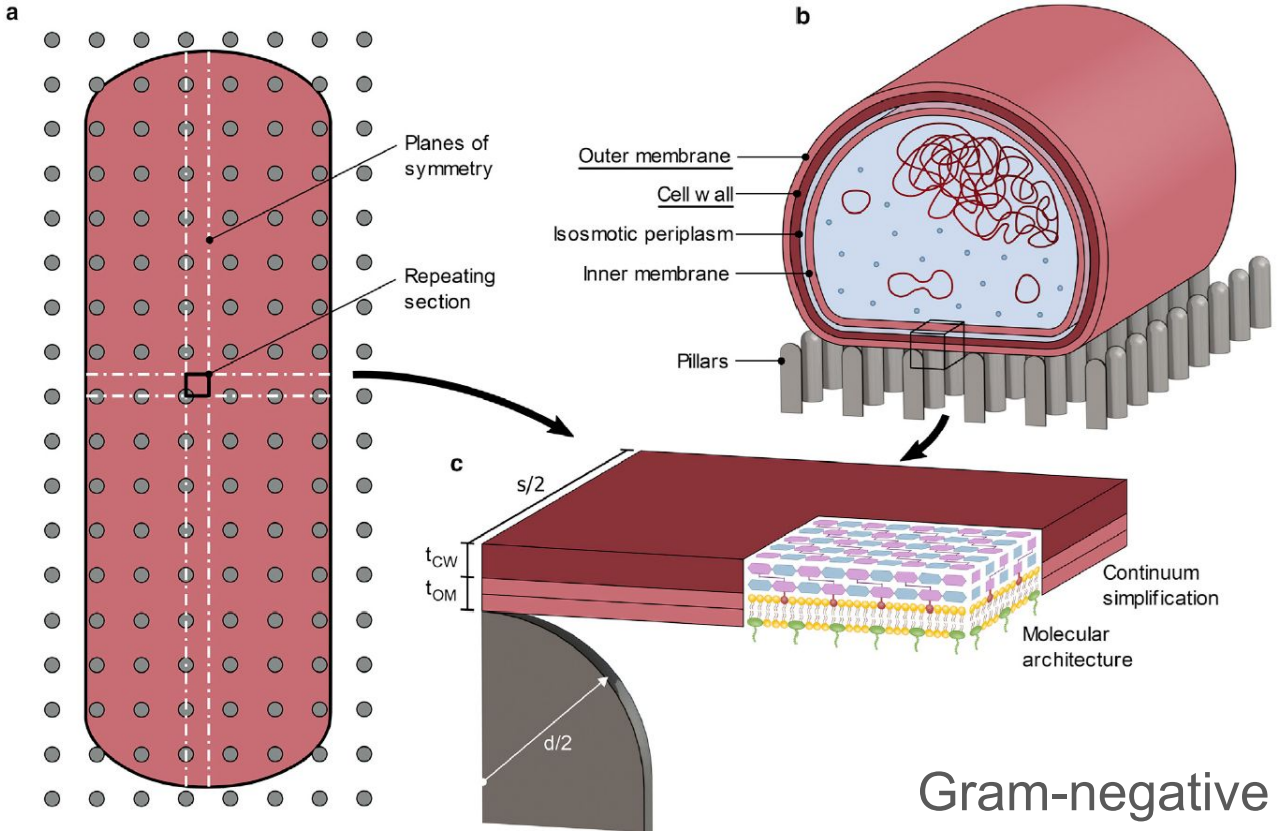
EQUILIBRIUM

4. Biophysical modelling: Velic et al. 2021

- a robust computational approach: FEM



4. Biophysical modelling: Velic et al. 2021

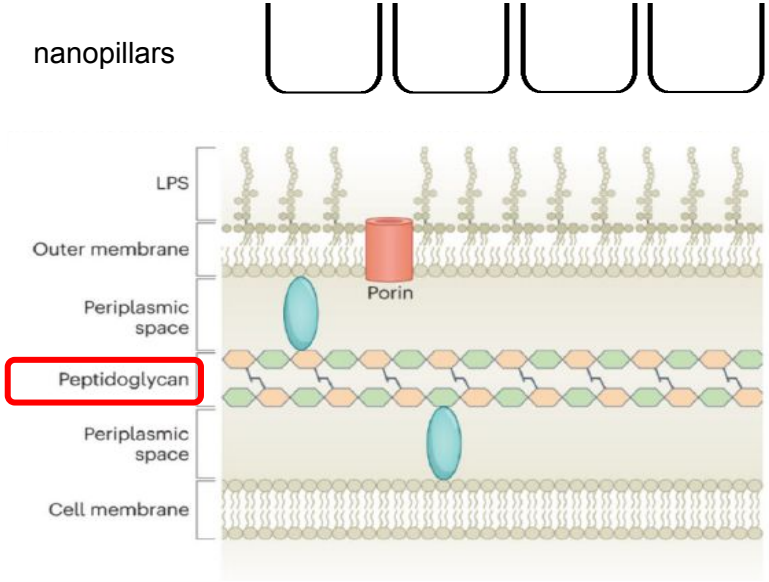


Gram-negative bacterium

4. Biophysical modelling: Velic et al. 2021

CELL WALL

- peptidoglycan → directional stiffness
- **linear orthotropic material**



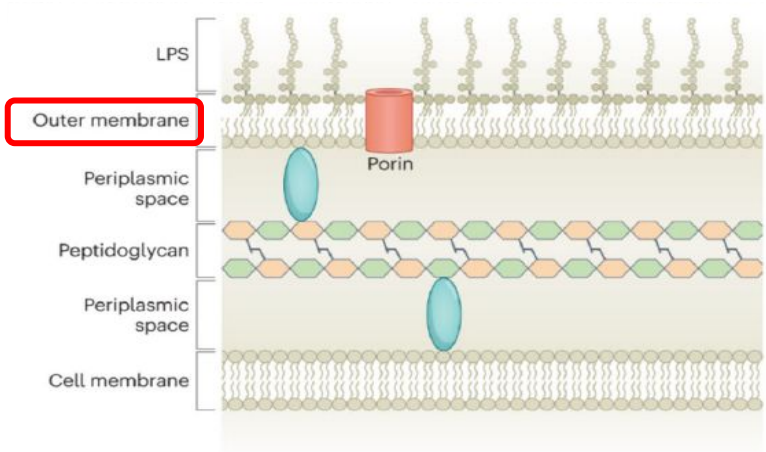
4. Biophysical modelling: Velic et al. 2021

CELL WALL

- linear orthotropic material

nanopillars

OUTER MEMBRANE



- phospholipid → strain-softening
- **isotropic neo-Hookean hyperelastic material**

$$D_1 = 2(C_{10} \cdot d)$$

4. Biophysical modelling: Velic et al. 2021

CELL WALL

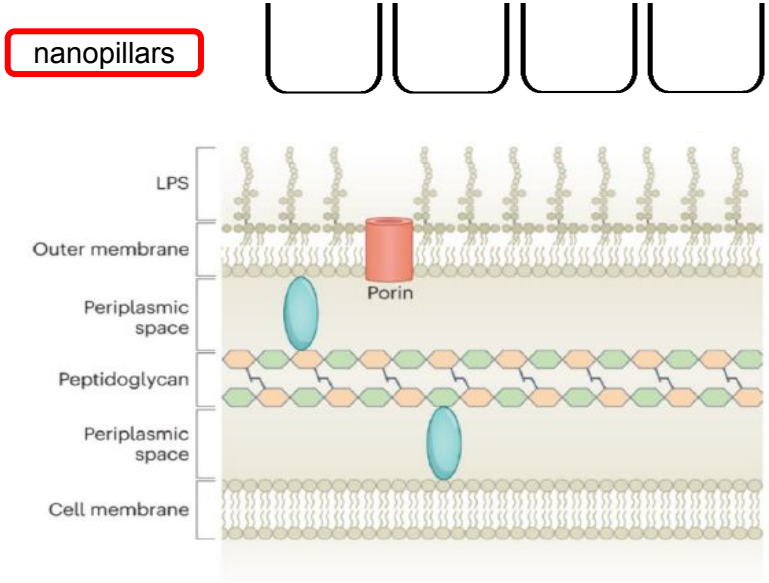
- linear orthotropic material

OUTER MEMBRANE

- isotropic neo-Hookean hyperelastic material

NANOPILLARS

- no deformity under bacterium
- **rigid** (infinitely stiff)



4. Biophysical modelling: Velic et al. 2021

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CELL WALL

- linear orthotropic material

OUTER MEMBRANE

- isotropic neo-Hookean hyperelastic material

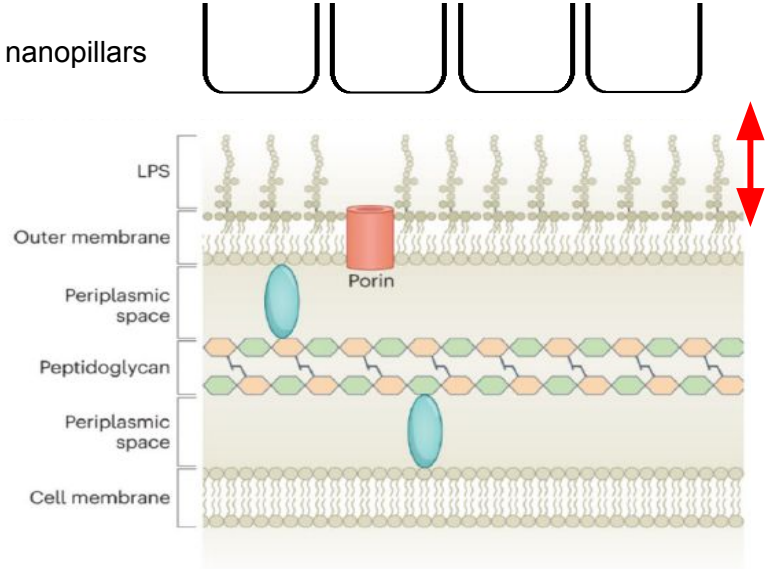
PILLARS

- rigid

MATERIAL PROPERTIES

CONTACT MODELS?

4. Biophysical modelling: Velic et al. 2021



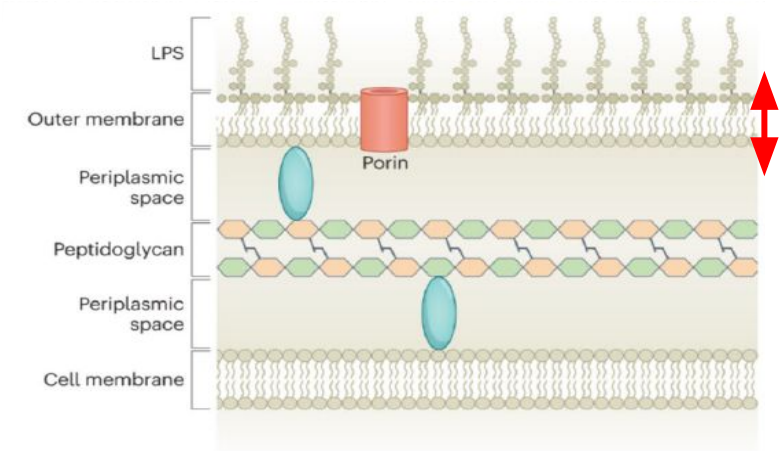
ADHESION

- **rough friction**
- once it touches, it sticks
- propagating bond front

$$W = \frac{\mathrm{d}U}{\mathrm{d}A}$$

4. Biophysical modelling: Velic et al. 2021

nanopillars



ADHESION

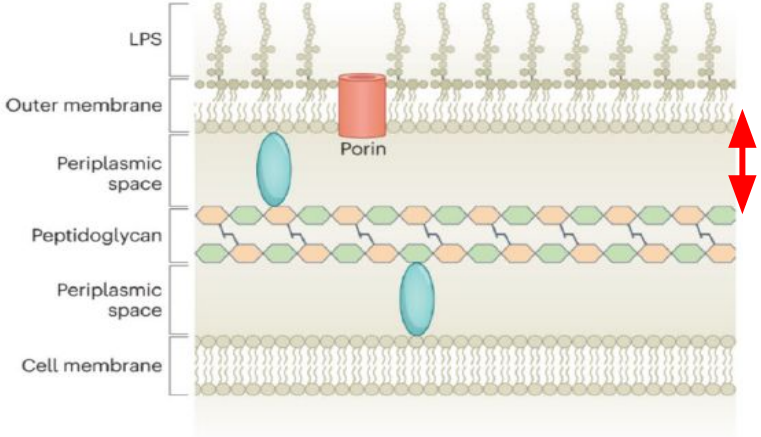
- rough friction

HYDROPHOBIC EFFECT

- **no friction**
- can slide off each other

4. Biophysical modelling: Velic et al. 2021

nanopillars



ADHESION - rough friction

HYDROPHOBIC EFFECT - no friction

INTERLAYER COUPLING

- covalent lipoprotein linkages
- **tie constraint**

4. Biophysical modelling: Velic et al. 2021

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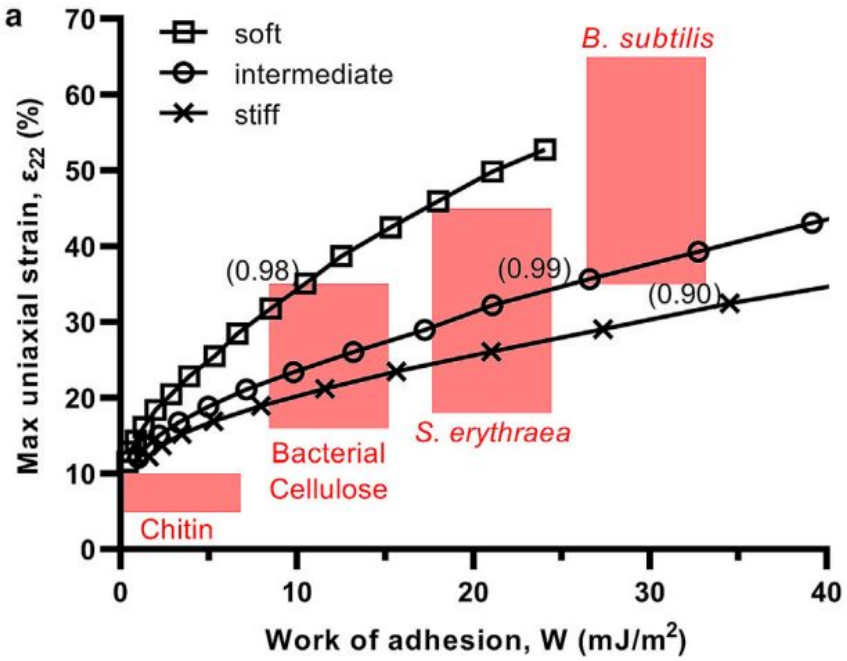
CELL WALL	- linear orthotropic material	**ADHESION**	- rough friction
OUTER MEMBRANE	- isotropic neo-Hookean hyperelastic material	**HYDROPHOBIC EFFECT**	- no friction
PILLARS	- rigid	**INTERLAYER COUPLING**	- tie constraint

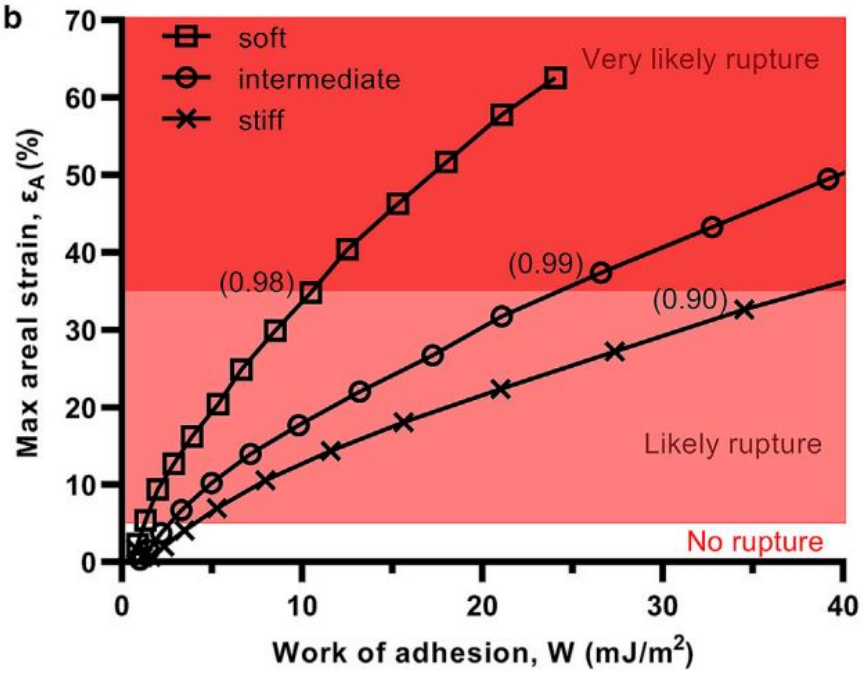
MATERIAL PROPERTIES

CONTACT MODELS

FEM

4. Biophysical modelling: Velic et al. 2021

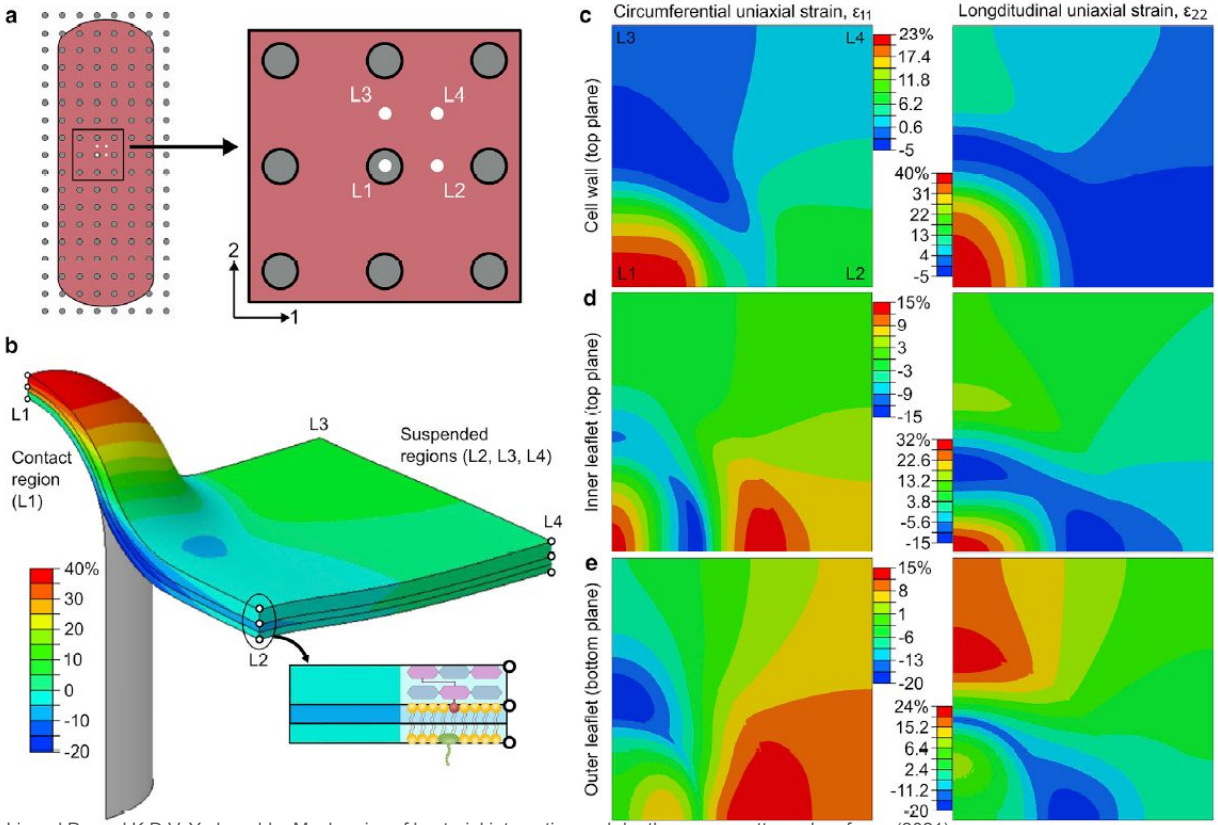




$$\varepsilon_A = (1+\varepsilon_{11})(1+\varepsilon_{22}) - 1$$

Amar Velic, Jafar Hasan, Zhiyong Li, and Prasad K.D.V. Yarlagadda. Mechan ics of bacterial interaction and death on nanopatterned surfaces (2021)

4. Biophysical modelling: Velic et al. 2021



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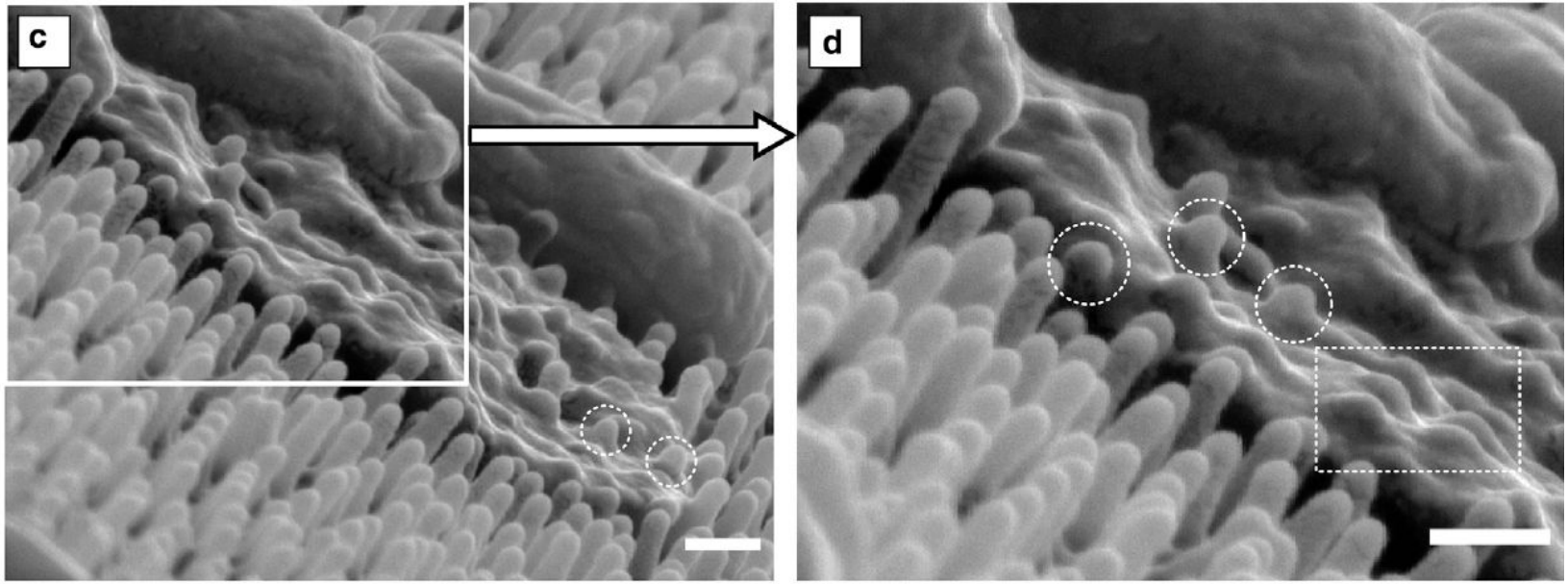
4. Biophysical modelling: Velic et al. 2021

VALIDATION - micrographs



Amar Velic, Jafar Hasan, Zhiyong Li, and Prasad K.D.V. Yarlagadda. Mechan ics of bacterial interaction and death on nanopatterned surfaces (2021)

5. Challenges

A. The "Elastic Balloon" Fallacy (Static vs. Dynamic Models)

B. The Pierce vs. Shear Debate

C. The Gram-Type Debate - Which bacteria are killed more efficiently?

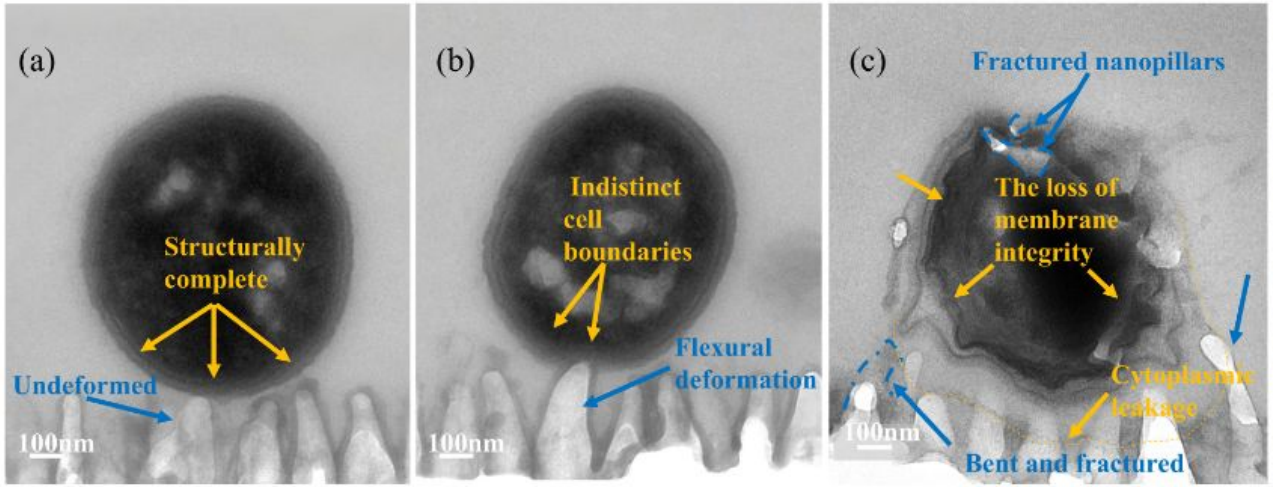
D. Beyond Pure Mechanics: Oxidative Stress

E. The Biofouling Problem: Cellular Debris

5. Challenges

Current State: We know nanostructures kill bacteria, but the exact physical mechanism is debated

TEM imaging of bacterial cells and nanopillar.
(a–c) Process of nanopillar–bacterium contact, deformation, and rupture



Qu, J., Gu, S., Chen, L., Cui, F., Wang, H., & Wu, L. (2026). Mechano-bactericidal activity of cicada wing nanostructures against gram-positive bacteria.

5. Challenges: C. The Gram-Type Debate

Which bacteria are killed more efficiently?

Gram-Type	Gram-positive	Gram-negative
Peptidoglycan layer	Thick (20-80 nm)	Thin (7-10 nm)
Rigidity	Higher ($E \approx 50-100$ MPa)	Lower rigidity ($E \approx 2-10$ MPa)
Preliminary Conclusions	Layer prevents critical membrane strain	Prone to severe deformation and tensile failure upon adhesion

Conclusion: Gram-negative bacteria are physically softer. They reach their elastic limit (Yield Point) and rupture much faster under capillary and adhesion forces

5. Challenges: C. The Gram-Type Debate

General trend: Gram-negative bacteria are **often more susceptible** to nanostructure-induced mechanical killing.

But: This trend is **not universal**.

Depending on nanostructure geometry: Gram-positive bacteria may show **similar or higher susceptibility**

5. Challenges: C. The Gram-Type Debate

A recent review summarizes:

Gram-positive bacteria are often more resilient, but many studies report equal or higher bactericidal efficiency depending on nanoscale architecture.

Bactericidal efficiency of micro- and nanostructured surfaces: a critical perspective

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Micro/nanostructured surfaces (MNSS) have shown the ability to inactivate bacterial cells by physical means. An enormous amount of research has been conducted in this area over the past decade. Here, we review the various surface factors that affect the bactericidal efficiency. For example, surface hydrophobicity of the substrate has been accepted to be influential on the bactericidal effect of the surface, but a review of the literature suggests that the influence of hydrophobicity differs with the bacterial species. Also, various bacterial viability quantification methods on MNSS are critically reviewed for their suitability for the purpose, and limitations of currently used protocols are discussed. Presently used static bacterial viability assays do not represent the conditions of which those surfaces could be applied. Such application conditions do have overlaying fluid flow, and bacterial behaviours are drastically different under flow conditions compared to under static conditions. Hence, it is proposed that the bactericidal effect should be assessed under relevant fluid flow conditions with factors such as shear stress and flowrate given due significance. This review will provide a range of opportunities for future research in design and engineering of micro/nanostructured surfaces with varying experimental conditions.

5. Challenges: D. Oxidative stress

The "Stretching vs. Piercing" debate **ignores cellular active responses**

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Antibacterial effects of nanopillar surfaces are mediated by cell impedance, penetration and induction of oxidative stress

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5. Challenges: D. Oxidative stress

Oxidative Stress (ROS) Induction:

Mechanical impedance on nanopillars triggers severe electrochemical reactions before physical rupture

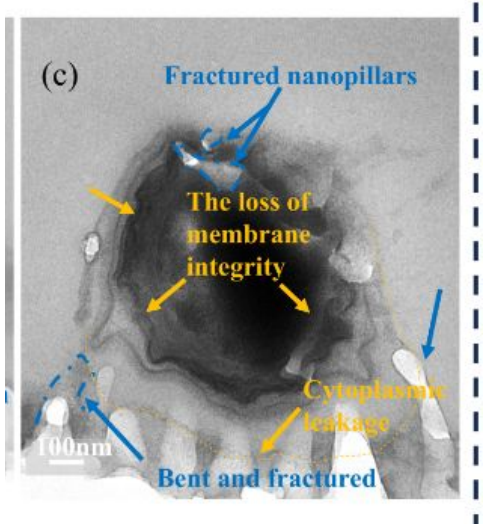
A Proposed Methodology to study oxidative stress:

- Couple **AFM (Atomic Force Microscopy)** with in-situ **Fluorescence Microscopy**
- Use ROS-sensitive dyes (e.g., DCFDA).
- Correlate physical indentation force (Nano-Newtons) with real-time ROS fluorescence intensity

5. Challenges: E. Biofouling problem

How do we deal with the toxic waste of dead bacteria?

Lysed cells leave debris → **"Conditioning film" forms** → **Nanotopography masked**



Loss of Topography: The sharp pillars become blunt, dropping bactericidal efficiency to near zero for subsequent colonizers

Toxicity: Accumulation of endotoxins triggers severe immune responses in clinical settings (implant rejection)

Qu, J., Gu, S., Chen, L., Cui, F., Wang, H., & Wu, L. (2026). Mechano-bactericidal activity of cicada wing nanostructures against gram-positive bacteria.

Engineering Solutions to Debris Clearance

Solution 1: Superhydrophobicity (Cassie-Baxter State)

- Designing the nanopillars to maintain a contact angle > 150º
- Fluid shear stress (e.g., blood flow or water flushing) naturally washes away the cellular debris (Self-cleaning effect)

Solution 2: Stimuli-Responsive Polymers (Smart Materials)

- Coating structures with polymers (e.g., PNIPAM) that change conformation (swell/collapse) with temperature shifts, physically "kicking off" the dead cells.

6. Outlook

- Integrating **fluid dynamics** and **smart materials** is mandatory for long-term real-world applications
- Future research must use **Machine Learning** to optimize pillar geometry (height, pitch, stiffness) against specific pathogen mechanical profiles
- Simulating fluid dynamics and cell-surface interactions in-silico will allow us to design smart, self-cleaning packaging and selective medical implants without relying on trial-and-error

7. Conclusion

- Nanotopography induces cell lysis via physical stress, bypassing chemical antimicrobial resistance
- Topographical bactericidal action is a validated, mechanics-driven phenomenon, highly dependent on the cell wall's Young's Modulus
- Gram-negative bacteria are mechanically more vulnerable, but surface fouling by cell debris remains a critical hurdle
- **Paradigm Shift:** Purely static and mechanical models (Hertz, JKR) must evolve into dynamic, multiphysics models that account for other forces and ROS generation

Bacterial-nanostructure interactions

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Questions?